# ARTIFICIAL INTELLIGENCE - PHASE 2

# CREATE A CHATBOT IN PYTHON

## PROBLEM STATEMENT:

The challenge is to create a Chatbot in Python that provides exceptional customer service, answering user queries on a website or application.

**Objectives:**

- Deliver high quality support to the users
- Ensuring a positive user experience
- Customer satisfaction

## SOLUTION:

Python AI chatbots are designed to simulate human-like conversation using Natural Language Processing (NLP) and Machine Learning.

These chatbots are capable of understanding and responding to text or voice inputs in natural language, providing seamless customer service, answering queries, or even making product recommendations.

### Exploring Natural Language Processing (NLP) in Python

Natural Language Processing, often abbreviated as NLP, is the cornerstone of any intelligent chatbot. NLP is a subfield of AI that focuses on the interaction between humans and computers using natural language.

## STEPS TO BE FOLLOWED:

**Step 1: Install and import Required Libraries**

Install the ChatterBot library using pip to get started on your chatbot journey.

pip install chatterbot

Import ChatterBot and its corpus trainer to set up and train the chatbot.

**Step 2: Train Your Chatbot with a Predefined Corpus**

Use the ChatterBotCorpusTrainer to train your chatbot using an English language corpus.

Train the bot with the custom corpus that was created. The bot can be trained with multiple files within a folder. For example, we have a folder with various YAML files.

**Step 3: Test Your Chatbot**

Interact with your chatbot by requesting a response to a greeting.

**Understanding User Intent**

Set up the testing environment utilize NLP techniques like Named Entity Recognition (NER) and Intent Classification to interpret user input.

**Step 4: Train Your Chatbot with Custom Data**

Make your chatbot more specific by training it with a list of your custom responses. Leverage machine learning models trained on large datasets to better recognize and respond to varied user queries.

Train your chatbot to handle unfamiliar queries gracefully. This could involve directing users to human support or suggesting alternate queries. Additionally, incorporate regular updates and training to your chatbot based on new and trending queries.

### Step 5: Integrate Your Chatbot into a Web Application

Use Flask to create a web interface for your chatbot, allowing users to interact with it through a browser.

- install the plugin, you're most like to redirect to the provider's website to Create an account.
- Once created, Create your own Chatbot or use templates.
- Check on your website, if the Chatbot is running as set up.

## FLOWCHART: